PRESIDENT'S OFFILE

REGIONAL ADMINISTRATION AND LOCAL GOVERMENT

ARUSHA REGION MOCK EXAMINATION

CHEMISTRY 2A MARKING GUIDE

CamScanner

TABLE OF RESULTS

EXPERIMENT NO.	PILOT	1	2	3
Final volume (cm^3)	20.20	40.20	20.20	40.10
Initial volume (cm^3)	0.00	20.20	0.10	20.20
Volume used (cm^3)	20.20	20.00	20.10	19.90

T = 02
D = 01
A = 02
(5)

Volume of pipette used = $20cm^3$ 01

Average volume = $\left(\frac{V_1 + V_2 + V_3}{3}\right) cm^3$

$= \frac{20.00 + 20.10 + 1990}{3}$

∴ Average volume = $20.00cm^3$ 01

a) $20.00cm^3$ of acid R was required to neutralize $20.00\ cm^3$ of the base S 01

b) Yellow to red/Pink/orange 01

c) Conc. of R in mol/dm^3

Conc. of R = $\frac{mass\ (g)}{Volume\ (dm^3)}$ 01

$= \frac{3.65g}{1dm^3}$

$= 3.65\ g/dm^3$ 01

Then molarity = $\frac{Conc.\ (g/dm^3)}{Molar\ mass}$

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Then molarity = $\frac{Conc.\ (g/dm^3)}{Molar\ mass}$

$$= \frac{3.65\ g/dm^3}{36.5\ g/mol}$$

$$= 0.1\ mol//dm^3$$ 01

d) $X_2CO_{3(aq)} + 2HCL_{(aq)} \rightarrow 2XCL_{(aq)} + CO_{2(g)} + H_2O_{(l)}$ 1 Mark

IONIC EQUATIONS

$X_2CO_{3(aq)} + 2HCL_{(aq)} \longrightarrow 2XCL_{(aq)} + CO_{2(g)} + H_2O_{(l)}$ ½

$2X^+_{(aq)}CO^{2-}_{3(aq)} + 2H^+_{(aq)}CL^-_{(aq)} \rightarrow 2X^+_{(aq)}2CL^-_{aq} + CO_{2(g)} + H_2O_{(l)}$ ½

$$CO^{2-}_{3(aq)} + 2H^+_{(aq)} \longrightarrow CO_{2(g)} + H_2O_{(l)}$$ 01

2

e) Data given

Molarity of acid $(Ma) = 0.1\ m$

Volume of acid $(Va) = 20.00 cm^3$

Volume of base $(Vb) = 20.00\ cm^3$

Number of moles of acid $(na) = 2$ 1 Mark

Number of moles of base $(nb) = 1$

Molarity of base $(Mb) = ?$

from

$$\frac{MaVa}{MbVb} = \frac{na}{nb}$$

$$M_b = \frac{M_a V_a n_b}{V_b n_a} \quad \text{---} \quad 0\frac{1}{}$$

$$= \frac{0.\frac{1}{}M \times 20.00cm^3 \times \frac{1}{}}{20.00cm^3 \times 2} \quad 0\frac{1}{}$$

$$= 0.05M$$

∴ Molarity of base = 0.05 M 0$\frac{1}{}$

Then

From concentration = $\frac{\text{mass (g)}}{\text{Volume } (dm^3)}$ of Malic

$$= \frac{2.65g}{0.5\,dm^3}$$

Concentration of $X_2CO_3 = 5.3g/dm^3$ 0$\frac{1}{}$

$$\text{Molarity} = \frac{\text{Conc. } (g/dm^3)}{\text{Molar mass } (g/mol)}$$

$$= \frac{5.3\,g/dm^3}{0.05\,mol/dm^3}$$

$$= 106\,g/mol \; 0\frac{1}{}$$

∴ The molar mass of $X_2CO_3 = 106g/mol$

Then $X_2CO_3 = 106$

$$2x + 12 + (16 \times 3) = 106 \quad \text{01}$$

$$2x + 60 = 106$$

$$\frac{2x}{2} = \frac{46}{2}$$

$$X = 23$$

Atomic mass of element X = 23 01

Element X is Sodium (Na) 01

r2

a) EFFECTS OF CONCENTRATION ON THE RATE OF CHEMICAL REACTION. 2 marks

b) TABLE OF RESULTS

EXPERIMENT	E (cm^3)	F (cm^3)	G (cm^3)	Time (S)	1/t (sec^{-1})
1	10	20	0	38	0.03
2	10	15	5	50	0.02
3	10	10	10	88	0.01
4	10	5	15	186	0.005

8 marks

C) 1/t represents the rate of chemical reactions — 1Mark
~~2 marks~~

D) $Na_2S_2O_{3(aq)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H_2O_{(l)} + S_{(s)} + SO_{2(g)}$ 1Mark

$Na^+_{(aq)} S_2O^{2-}_{3(aq)} + 2H^+_{(aq)} 2Cl^-_{(aq)} \rightarrow 2Na^+ 2Cl^-_{(aq)} + H_2O_{2(l)} + S_{(s)} + SO_{2(g)}$ 1Mark

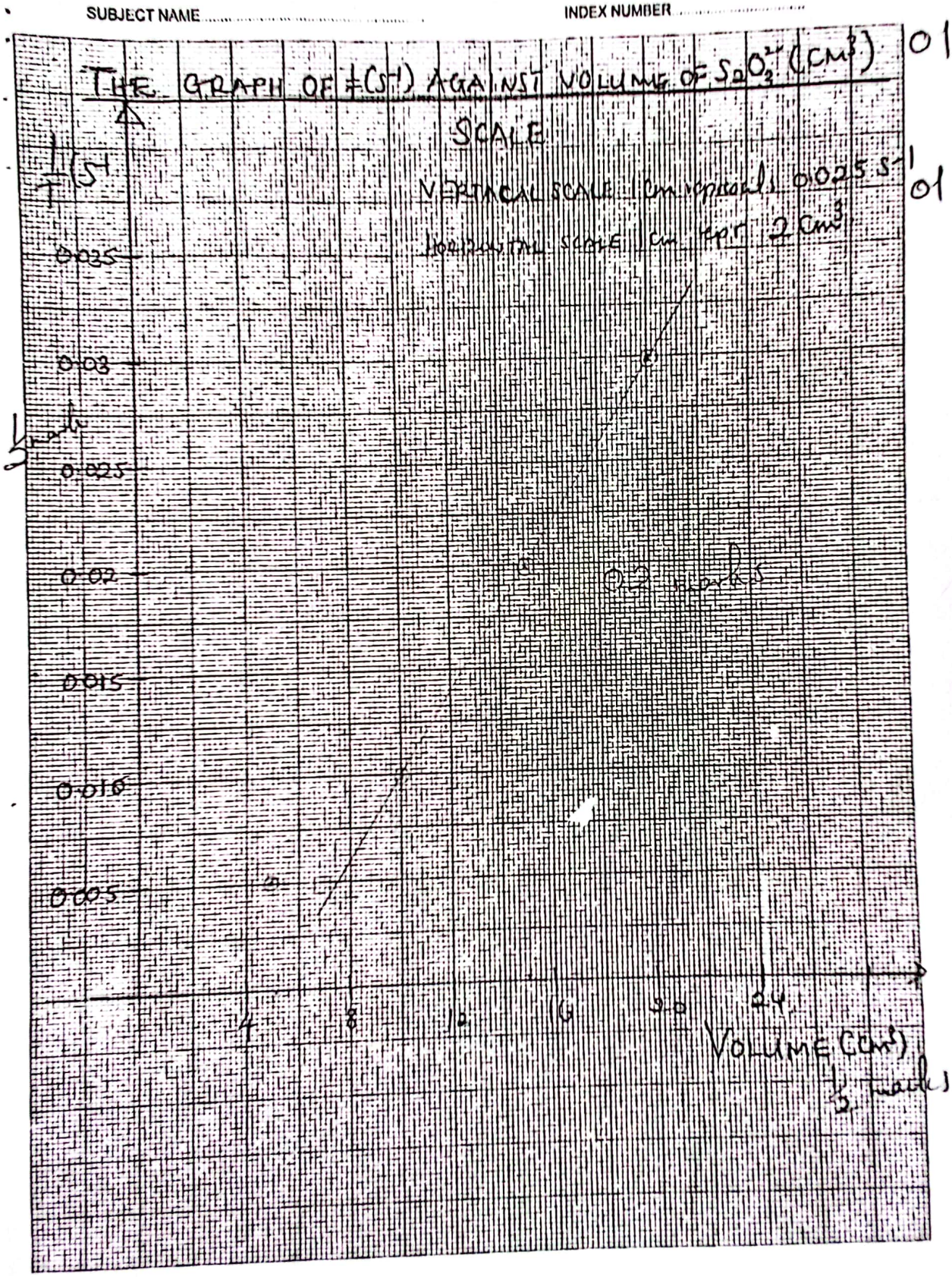
$S_2O_3^{2-}{}_{(aq)} + 2H^+_{(aq)} \longrightarrow S_{(s)} + SO_{2(g)} + H_2O_{(L)}$ 1M
~~2 marks~~

E) Sulphur precipitates 2 marks

F) i) Manufacture sulphuric acid
ii) Vulcanization of rubber
iii) Manufacture of gunpowder 2 marks
iv) Manufacture of medicine etc

H) Increase the concentration increase the rate of chemical reaction. 2 marks

SUBJECT NAME ..... INDEX NUMBER .....



0/mark

0/mark

½ mark